

Intel® Distribution for GDB* Cheat Sheet

Prerequisites

Set your oneAPI environment variables:

```
$ source <ONEAPI_ROOT>/setvars.sh
```

For GPU debugging:

Enable debug support in the kernel by setting the debug-enablement sysfs entry to 1:

```
find /sys/class/drm/card*/device/ \
-name "*enable_eu*debug" \
-exec sh -c 'echo 1 | sudo tee "$1"' _ {} \;
```

Define the following environment variable:

```
$ export ZET_ENABLE_PROGRAM_DEBUGGING=1
```

Check that your user is a member of the linux group that owns the graphics card. [↗](#)

Useful GDB Commands

help <cmd>

Print help info about the command *cmd*.

run [arg1, ... argN]

Start the program, optionally with arguments.

break <filename>:<line>

Define a breakpoint at given source file's specified line.

info break

Show the defined breakpoints.

delete <N>

Remove the Nth breakpoint.

watch <exp>

Stop when value of the expression *exp* changes. Not supported for GPU debugging.

step, next

Single-step a source line, stepping into/over func calls.

continue

Continue execution.

print <exp>

Print value of expression *exp*.

backtrace

Show the function call stack.

up, down

Go one level up/down in the function call stack.

disassemble

Disassemble the current function.

info args/locals

Show the arguments/local vars of the current function.

info reg <regname>

Show contents of the specified register.

info inferiors

Display information about the *inferiors*. For GPU offloading, the host process and the GPU devices are each represented by an inferior.

info devices

Display information about the *devices*.

Useful GDB Commands (cont'd)

info threads [-stopped] <ID>

Display information about threads with id *ID*, including their active SIMD lanes. Omit id to display all threads.

Use the *-stopped* flag to limit to stopped threads.

thread <thread_id>:<lane>

Switch context to the SIMD lane *lane* of the specified thread. E.g: `thread 2.6:4`

threads apply [-unavailable] [-c] [-q] [-s]

all | all-lanes | <ID>:<lane> <cmd>

Apply command *cmd* to the specified threads and lanes.

Useful for inspecting vectorized values.

thread filter <ID>:<lane> | all | all-lanes

<-location loc <expression>

Filter from the specified range of threads using the expression and the location options.

x /<format> <addr>

Examine the memory at address *addr* according to *format*. E.g: 'x /i \$pc' shows the instruction pointed by the program counter. 'x /8wd &count' shows 8 words in decimal format located at the address of *count*.

set nonstop on/off

Enable/disable the nonstop mode. This command may *not* be used after the program has started.

set scheduler-locking continue/eval/step/replay on/off

Control the thread scheduler. Useful to keep the other threads stopped while the current thread is stepping (if *step* is on) or resumed (if *continue* is on) to avoid interference.

maint jit dump <addr> <filename>

Save the JIT'ed objfile that contains address *addr* into the file *filename*. Useful for extracting the SYCL kernel when running on the CPU device.

cond [-force] <N> <exp>

Define the expression *exp* as the condition for breakpoint *N*. Use the optional *-force* flag to force the condition to be defined even when *exp* is invalid for the current locations of the breakpoint. Useful for defining conditions on breakpoints in JIT-produced code.

Notes for GPU debugging

Currently only the Intel® oneAPI Level Zero backend is supported for GPU debugging.

Workloads submitted to different devices and/or subdevices can be debugged simultaneously. Only one workload at a time can be debugged on a subdevice. Other workloads submitted to the same subdevice need to wait until the subdevice is free again.

The `ZE_AFFINITY_MASK=<device>.<subdevice>` environment variable [↗](#) can be used to limit the devices/subdevices available to the program.

Links

[oneAPI Programming Guide](#) [↗](#) [Get Started Guide](#) [↗](#)
[Release Notes](#) [↗](#)

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